Algebra II with Trigonometry Chapter 7.3

Olympiad Solutions (Gemini)

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Solutions

1. Evaluate the product
$32\cos\left(\frac{\pi}{33}\right)\cos\left(\frac{2\pi}{33}\right)\cos\left(\frac{4\pi}{33}\right)\cos\left(\frac{8\pi}{33}\right)\cos\left(\frac{16\pi}{33}\right)$.

Solution: Let P be the given product. We multiply and divide by $\sin\left(\frac{\pi}{33}\right)$ and repeatedly apply the double-angle identity $\sin(2\theta) = 2\sin\theta\cos\theta$:

$$\begin{aligned} P &= \frac{16\left[2\sin\left(\frac{\pi}{33}\right)\cos\left(\frac{\pi}{33}\right)\right]\cos\left(\frac{2\pi}{33}\right)\cos\left(\frac{4\pi}{33}\right)\cos\left(\frac{8\pi}{33}\right)\cos\left(\frac{16\pi}{33}\right)}{\sin\left(\frac{\pi}{33}\right)} \\ &= \frac{8\left[2\sin\left(\frac{2\pi}{33}\right)\cos\left(\frac{2\pi}{33}\right)\right]\cos\left(\frac{4\pi}{33}\right)\cos\left(\frac{8\pi}{33}\right)\cos\left(\frac{16\pi}{33}\right)}{\sin\left(\frac{\pi}{33}\right)} \\ &= \frac{4\sin\left(\frac{4\pi}{33}\right)\cos\left(\frac{4\pi}{33}\right)\dots}{\sin\left(\frac{\pi}{33}\right)} \\ &= \dots = \frac{\sin\left(\frac{32\pi}{33}\right)}{\sin\left(\frac{\pi}{33}\right)} \end{aligned}$$

Because $\frac{32\pi}{33} = \pi - \frac{\pi}{33}$, we have $\sin\left(\frac{32\pi}{33}\right) = \sin\left(\frac{\pi}{33}\right)$. Thus, the numerator and denominator cancel perfectly, yielding:

$$P = 1$$

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2. Find the exact value of $\sin 12^\circ + \sin 48^\circ - \sin 72^\circ$.

Solution: We group the second and third terms and apply the sum-to-product identity $\sin x - \sin y = 2\sin\left(\frac{x-y}{2}\right)\cos\left(\frac{x+y}{2}\right)$:

$$\begin{aligned}\sin 12^\circ + (\sin 48^\circ - \sin 72^\circ) &= \sin 12^\circ + 2\sin\left(\frac{48^\circ - 72^\circ}{2}\right)\cos\left(\frac{48^\circ + 72^\circ}{2}\right) \\ &= \sin 12^\circ + 2\sin(-12^\circ)\cos(60^\circ)\end{aligned}$$

Since $\sin(-12^\circ) = -\sin 12^\circ$ and $\cos(60^\circ) = \frac{1}{2}$, this simplifies to:

$$\sin 12^\circ - 2\sin 12^\circ\left(\frac{1}{2}\right) = \sin 12^\circ - \sin 12^\circ = 0$$

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3. Evaluate $\csc\left(\frac{\pi}{9}\right)\sqrt{2 - \sqrt{2 + \sqrt{2 + 2\cos\left(\frac{8\pi}{9}\right)}}}$**.**

Solution: We collapse the nested radicals from the inside out using the half-angle identities $1 + \cos(2\theta) = 2\cos^2\theta$ and $1 - \cos(2\theta) = 2\sin^2\theta$. For the innermost radical:

$$2 + 2\cos\left(\frac{8\pi}{9}\right) = 4\cos^2\left(\frac{4\pi}{9}\right)$$

Because $\frac{4\pi}{9} \in \left[0, \frac{\pi}{2}\right]$, its cosine is positive, making the square root exactly $2\cos\left(\frac{4\pi}{9}\right)$. Substituting this into the next radical:

$$\sqrt{2 + 2\cos\left(\frac{4\pi}{9}\right)} = \sqrt{4\cos^2\left(\frac{2\pi}{9}\right)} = 2\cos\left(\frac{2\pi}{9}\right)$$

Substituting this into the outermost radical (noticing the minus sign):

$$\sqrt{2 - 2\cos\left(\frac{2\pi}{9}\right)} = \sqrt{4\sin^2\left(\frac{\pi}{9}\right)} = 2\sin\left(\frac{\pi}{9}\right)$$

Finally, we multiply this result by the coefficient in front:

$$\csc\left(\frac{\pi}{9}\right) \times 2\sin\left(\frac{\pi}{9}\right) = \frac{1}{\sin\left(\frac{\pi}{9}\right)} \times 2\sin\left(\frac{\pi}{9}\right) = 2$$

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4. Find the exact value of $\tan\left(2\tan^{-1}(2) - \sin^{-1}\left(\frac{4}{5}\right)\right)$.

Solution: Let $\alpha = \tan^{-1}(2)$ and $\beta = \sin^{-1}\left(\frac{4}{5}\right)$. Since $\tan\alpha = 2 > 1$, we have $\alpha \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$, so $2\alpha \in \left(\frac{\pi}{2}, \pi\right)$. We compute the sine and cosine of 2α:

$$\sin(2\alpha) = \frac{2\tan\alpha}{1+\tan^2\alpha} = \frac{4}{5}, \quad \cos(2\alpha) = \frac{1-\tan^2\alpha}{1+\tan^2\alpha} = -\frac{3}{5}$$

From $\beta = \sin^{-1}\left(\frac{4}{5}\right)$, we know $\sin\beta = \frac{4}{5}$ and, given $\beta \in \left[0, \frac{\pi}{2}\right]$, $\cos\beta = \frac{3}{5}$. Notice that $\sin(2\alpha) = \sin\beta$ and $\cos(2\alpha) = -\cos\beta$. Since 2α is in the second quadrant and β is in the first, they represent supplementary angles. Hence, $2\alpha = \pi - \beta$. We are tasked with evaluating $\tan(2\alpha - \beta)$:

$$\tan(2\alpha - \beta) = \tan(\pi - 2\beta) = -\tan(2\beta)$$

Using $\tan\beta = \frac{\sin\beta}{\cos\beta} = \frac{4}{3}$, we apply the double-angle tangent formula:

$$-\tan(2\beta) = -\frac{2\tan\beta}{1-\tan^2\beta} = -\frac{2\left(\frac{4}{3}\right)}{1-\left(\frac{4}{3}\right)^2} = -\frac{\frac{8}{3}}{1-\frac{16}{9}} = -\frac{\frac{8}{3}}{-\frac{7}{9}} = \frac{24}{7}$$

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5. Find the exact value of the sum $\sum_{k=1}^{6} \sin^4\left(\frac{(2k-1)\pi}{12}\right)$.

Solution: The angles in the sum are $\frac{\pi}{12}, \frac{3\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{9\pi}{12}, \frac{11\pi}{12}$. Because $\sin(\pi - \theta) = \sin\theta$, the terms for $k = 4, 5, 6$ mirror those for $k = 3, 2, 1$. Therefore, the sum simplifies to:

$$S = 2\left[\sin^4\left(\frac{\pi}{12}\right) + \sin^4\left(\frac{3\pi}{12}\right) + \sin^4\left(\frac{5\pi}{12}\right)\right]$$

Notice that $\frac{3\pi}{12} = \frac{\pi}{4}$, so $\sin^4\left(\frac{3\pi}{12}\right) = \left(\frac{\sqrt{2}}{2}\right)^4 = \frac{1}{4}$. Also, $\frac{5\pi}{12} = \frac{\pi}{2} - \frac{\pi}{12}$, which means $\sin\left(\frac{5\pi}{12}\right) = \cos\left(\frac{\pi}{12}\right)$. Thus, the remaining two terms pair up elegantly:

$$\sin^4\left(\frac{\pi}{12}\right) + \cos^4\left(\frac{\pi}{12}\right) = \left(\sin^2\left(\frac{\pi}{12}\right) + \cos^2\left(\frac{\pi}{12}\right)\right)^2 - 2\sin^2\left(\frac{\pi}{12}\right)\cos^2\left(\right.$$

$$= 1 - \frac{1}{2}\sin^2\left(\frac{\pi}{6}\right) = 1 - \frac{1}{2}\left(\frac{1}{4}\right) = \frac{7}{8}$$

Substituting these parts back into the overall sum:

$$S = 2\left[\frac{7}{8} + \frac{1}{4}\right] = 2\left(\frac{9}{8}\right) = \frac{9}{4}$$

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6. Evaluate $\dfrac{\sin 20^\circ + \sin 40^\circ + \sin 60^\circ + \sin 80^\circ + \sin 100^\circ}{\cos 20^\circ + \cos 40^\circ + \cos 60^\circ + \cos 80^\circ + \cos 100^\circ}$**.**

Solution: Let N be the numerator and D be the denominator. We strategically pair the symmetrically spaced terms around the center angle of 60°.

For the numerator, we apply the sum-to-product identity $\sin x + \sin y = 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$:

$$\begin{aligned} N &= (\sin 20^\circ + \sin 100^\circ) + (\sin 40^\circ + \sin 80^\circ) + \sin 60^\circ \\ &= 2\sin(60^\circ)\cos(-40^\circ) + 2\sin(60^\circ)\cos(-20^\circ) + \sin 60^\circ \\ &= \sin 60^\circ\big(2\cos 40^\circ + 2\cos 20^\circ + 1\big) \end{aligned}$$

For the denominator, we analogously apply $\cos x + \cos y = 2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$:

$$\begin{aligned} D &= (\cos 20^\circ + \cos 100^\circ) + (\cos 40^\circ + \cos 80^\circ) + \cos 60^\circ \\ &= 2\cos(60^\circ)\cos(-40^\circ) + 2\cos(60^\circ)\cos(-20^\circ) + \cos 60^\circ \\ &= \cos 60^\circ\big(2\cos 40^\circ + 2\cos 20^\circ + 1\big) \end{aligned}$$

When we divide N by D, the non-zero parenthetical factor drops out entirely, yielding:

$$\frac{N}{D} = \frac{\sin 60^\circ \left(2\cos 40^\circ + 2\cos 20^\circ + 1\right)}{\cos 60^\circ \left(2\cos 40^\circ + 2\cos 20^\circ + 1\right)} = \tan 60^\circ = \sqrt{3}$$